MACHINE LEARNING COURSE PROJECT | FINAL TECHNICAL REPORT

Industrial Machine Condition Classification

Multiclass classification for condition-based predictive maintenance

Task: 3-class condition classification
Validation: 5-fold stratified CV
Classes: Normal, Thermal Failure, Mechanical Failure
Training / test: 16,000 / 4,000 rows
Best reported OOF accuracy: 89.28%
Selected approach: Calibrated soft-voting ensemble + TTA

16,000	3	89.28%
Labeled training records	Target classes	Best reported OOF accuracy

Executive Summary

This project frames industrial machine condition monitoring as a three-class supervised learning problem. The training set contains 16,000 labeled observations: 8,000 Normal records and 4,000 records for each failure mode. Sensor measurements describe thermal conditions, cooling, rotational behavior, mechanical load, wear, and vibration.

The reported best model is a calibrated soft-voting ensemble of CatBoost, XGBoost, and LightGBM with test-time augmentation (TTA). It reports 89.28% out-of-fold (OOF) accuracy, 87.40% balanced accuracy, and 88.32% Macro F1. XGBoost is the strongest reported single-model alternative at 89.20% OOF accuracy.

Item	Details
Problem	3-class classification: Normal, Thermal Failure, Mechanical Failure
Data	16,000 labeled training rows and 4,000 test rows
Validation	5-fold Stratified Cross-Validation
Best reported ensemble	CatBoost 35.31%, XGBoost 32.73%, LightGBM 31.96%
Best reported metrics	89.28% accuracy; 87.40% balanced accuracy; 88.32% Macro F1

Results and metrics are reproduced from the supplied report. Public leaderboard results can vary and should be distinguished from cross-validation estimates.

1. Problem and Dataset

The objective is to classify the operational state of industrial machinery from multivariate sensor telemetry. Unexpected equipment failures can cause financial loss, safety risks, and production downtime. Condition-based predictive maintenance aims to identify degradation early enough to support timely maintenance.

Class	Condition	Records	Share
0	Normal	8,000	50.00%
1	Thermal Failure	4,000	25.00%
2	Mechanical Failure	4,000	25.00%

- **Files supplied:** student_dataset.csv (16,000 labeled observations), test.csv (4,000 unseen observations), and sample_submission.csv (unit_id, condition).
- **Input attributes:** unit_id identifier; categorical unit_type; thermal sensors ambient_temp, process_temp, coolant_flow, and thermal_stress; and mechanical/kinematic sensors shaft_speed, drive_torque, power_load, component_wear, mechanical_load, and vibration_index. The identifier is excluded from model training.

2. Exploratory Data Analysis (EDA)

- **Class balance and metrics:** Normal observations account for half the dataset. Accuracy alone may obscure errors in the two failure classes, so balanced accuracy, Macro F1, and Cohen's Kappa are also reported.
- **Thermal gradient:** Ambient and process temperatures have a reported correlation of approximately r = 0.61. Normal machines typically show a process-to-ambient temperature difference of about 10-16°C; thermal failures are associated with larger differences and cooling deficits.
- **Torque-speed relationship:** The report identifies a relationship between drive torque and shaft speed consistent with motor power behavior. Mechanical failures cluster in regions with elevated torque, component wear, and vibration.
- **Distinct failure patterns:** Thermal failure patterns are associated with thermal stress and coolant flow; mechanical failure patterns are associated with vibration, wear, and mechanical load.
- **Potential label noise:** The supplied report estimates that about 4% of records may contain contradictory labels from synthetic simulation noise. This is a dataset-specific observation, not a general property of industrial sensor data.

3. Data Preprocessing

The reported preprocessing workflow is modular and designed to avoid data leakage across cross-validation folds.

- **Missing values:** Numeric values were imputed with medians calculated on each fold's training partition. Missing unit_type values were imputed using the training mode, reported as M.
- **Categorical encoding:** Machine type was ordinal-mapped (S = 0, M = 1, H = 2); one-hot columns were generated for unit_type_S, unit_type_M, and unit_type_H.
- **Feature scaling:** StandardScaler was fitted within each fold for linear and generative models. Tree-based boosting models used unscaled features.
- **Physics-informed features:** The report states that 35 engineered features were created, including temperature difference, cooling ratio, speed-vibration interaction, torque-wear interaction, and rotational/thermal interactions.

- **Numerical safeguards:** Small epsilon values were added to denominators to reduce division-by-zero issues. Infinite values were converted to missing values and then imputed.
- **Down-weighting suspected noisy labels:** Out-of-fold probability checks reportedly identified 647 records with high model-label disagreement; those records received a sample weight of 0.25.

> **Preprocessing principle.** Imputation, scaling, and related transformations are fitted on the training portion of each fold before being applied to its validation portion. The report describes this as the mechanism used to prevent fold-level leakage.

4. Models Evaluated

The report compares linear, probabilistic, tree-ensemble, and gradient-boosting approaches. The table consolidates the reported out-of-fold metrics for the winning ensemble and candidate models.

Model	Accuracy	Balanced acc.	Macro F1	Log loss	Kappa	MCC
Calibrated Ensemble + TTA	89.28%	87.40%	88.32%	0.3112	0.8260	0.8273
XGBoost	89.20%	87.34%	88.24%	0.3125	0.8249	0.8261
Top-3 Equal Ensemble	89.18%	87.31%	88.21%	0.3105	0.8246	0.8258
CatBoost	89.16%	87.30%	88.17%	0.3126	0.8242	0.8254
LightGBM	88.99%	87.09%	88.00%	0.3143	0.8215	0.8227
Random Forest	88.69%	86.78%	87.62%	0.3329	0.8167	0.8178
Logistic Regression	85.79%	83.25%	84.18%	0.3935	0.7691	0.7706
Linear Discriminant Analysis	84.50%	81.21%	82.59%	0.4251	0.7460	0.7497
Quadratic Discriminant Analysis	83.88%	81.06%	81.97%	0.8523	0.7380	0.7402
Gaussian Naive Bayes	79.44%	76.67%	76.93%	1.9622	0.6692	0.6694

Model Comparison at a Glance

Model family	Best reported score	Interpretation
Calibrated ensemble + TTA	89.28% accuracy	Best reported combined performance
XGBoost	89.20% accuracy	Strongest reported individual model
Top-3 equal ensemble	89.18% accuracy	Equal blend of LightGBM, XGBoost, and CatBoost
CatBoost / LightGBM	89.16% / 88.99%	Other leading gradient-boosting models
Linear / probabilistic baselines	79.44-85.79%	Lower reported accuracy in this comparison

- **Gradient boosting leads:** XGBoost, CatBoost, and LightGBM are the strongest individual models in the reported comparison.
- **Naive Bayes trails:** Gaussian Naive Bayes has the lowest reported accuracy and highest log loss; correlated sensor features may violate its conditional-independence assumption.
- **LDA slightly exceeds QDA:** The report attributes this to pooled covariance estimation in LDA compared with separate class covariance matrices in QDA.

5. Hyperparameter Tuning

All models were evaluated using 5-fold Stratified Cross-Validation with shuffle=True and random_state=42, preserving class proportions across folds. The following configurations are reproduced from the report.

Model	Reported selected configuration
Logistic Regression	C = 1.0; lbfgs; max_iter = 1000
LDA	SVD solver
Gaussian Naive Bayes	var_smoothing = 1e-9
QDA	reg_param = 0.01
Random Forest	400 trees; max_depth = 19; min_samples_leaf = 2; max_features = 0.7
LightGBM	num_leaves = 48; max_depth = 7; learning_rate = 0.02; reg_alpha = 0.15; reg_lambda = 1.5
XGBoost	max_depth = 6; learning_rate = 0.02; subsample = 0.8; colsample_bytree = 0.8; min_child_weight = 4
CatBoost	depth = 7; learning_rate = 0.025; l2_leaf_reg = 5.0; random_strength = 0.8

Ensemble weights

The reported soft-voting blend assigns 35.31% to CatBoost, 32.73% to XGBoost, and 31.96% to LightGBM. Random Forest and baseline models were assigned zero weight in the optimized blend.

Model	Blend weight
CatBoost	35.31%
XGBoost	32.73%
LightGBM	31.96%

Class probability multipliers

The report gives multipliers from a grid search constrained by P(Normal) >= 50%: Normal = 1.000, Thermal = 0.9127, and Mechanical = 1.0200.

6. Results and Discussion

Reported ordering	Result
Calibrated Ensemble + TTA	89.28% accuracy; best reported score
XGBoost	89.20% accuracy; strongest single model
Top-3 equal ensemble	89.18% accuracy
CatBoost / LightGBM	89.16% / 88.99% accuracy
Random Forest	88.69% accuracy
Logistic / LDA / QDA / Gaussian NB	85.79% / 84.50% / 83.88% / 79.44%

Overall, the report attributes the stronger gradient-boosting results to their ability to model nonlinear decision boundaries and interactions between coupled industrial sensors. These results should be interpreted as dataset-specific experimental results.

7. Class Diagnostics

The confusion matrix below is reproduced from the supplied report for the winning calibrated ensemble. Rows are true labels and columns are predicted labels.

True class	Pred. Normal (0)	Pred. Thermal (1)	Pred. Mechanical (2)
True Normal (0)	7,592	224	184
True Thermal (1)	458	3,342	200
True Mechanical (2)	493	157	3,350

Class	Correctly classified	Recall	Precision
Normal	7,592 / 8,000	94.90%	88.87%
Thermal Failure	3,342 / 4,000	83.55%	89.77%
Mechanical Failure	3,350 / 4,000	83.75%	89.72%

The report notes that cross-confusion between thermal and mechanical failure is relatively limited compared with errors where failure observations are predicted as Normal. This highlights the practical importance of class-specific recall, not accuracy alone.

8. Sample Efficiency

The report compares full-data results with a randomly sampled 20% subset (2,560 records). The values below are reproduced as stated in the source and are separate from the OOF comparison table above.

Model	Full-data accuracy	20% subset accuracy	Change
Random Forest	89.31%	88.53%	-0.78 percentage points
Logistic Regression	86.41%	85.62%	-0.79 percentage points
XGBoost	90.06%	88.59%	-1.47 percentage points

According to the report, all three models retained more than 98% of their full-data accuracy in this subset experiment. The full-data values shown here are distinct from the earlier OOF table.

9. Best Model and Kaggle Submission

The selected model is the calibrated multi-model ensemble with test-time augmentation (TTA). The reported validation benchmark is 89.28% OOF accuracy, 87.40% balanced accuracy, 88.32% Macro F1, and 0.3112 log loss.

Validation metric	Reported value
OOF accuracy	89.28%
Balanced accuracy	87.40%
Macro F1	88.32%
Log loss	0.3112

- **Ensemble components:** CatBoost (35.31%), XGBoost (32.73%), and LightGBM (31.96%).
- **Single-model alternative:** XGBoost with 89.20% OOF accuracy and 88.24% Macro F1.
- **Submission format:** submission.csv contains 4,000 rows and uses columns unit_id and condition.
- **Reported predicted distribution:** 2,126 Normal (53.15%), 927 Thermal Failure (23.18%), and 947 Mechanical Failure (23.68%).

10. GitHub Repository Link

The project repository is available at the GitHub URL below.

PROJECT REPOSITORY

Industrial Machine Condition Classification

GitHub

https://github.com/PrafulPradhan/Industrial-Machine-Condition-Classification

Repository URL

Select the link above to open the project repository in your browser.

Conclusion

The reported results favor a blended gradient-boosting approach for this dataset. Key strengths include fold-aware preprocessing, physics-informed features, stratified validation, and class-sensitive evaluation. Scores are dataset-specific and do not guarantee performance on other industrial systems.